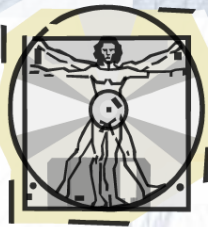


Assignment 4

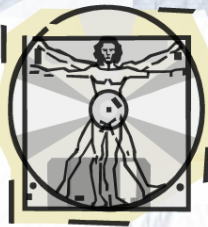




Assignment 4

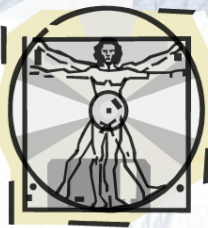
- Mutation Testing (Fault Injection)
- GUI/Web Test Automation





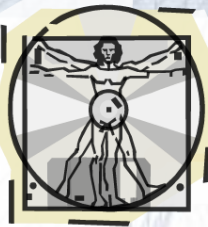
Activities

- 1st week:
 - Install and get familiar with a mutation test tool (e.g. muJava)
 - Install and get familiar with a GUI/Web test automation tool (e.g. Selenium, SikuliX, etc.)
- 2nd week:
 - Design and run mutation testing on your test suite
 - Design test cases (scripts) for some functions using test automation tool
 - Demo both parts



Assignment 4 – Part 1

- Install a mutation testing tool
- Run mutation testing for the given SUT and your own tests, and record the results
- Identify some equivalent mutants and describe your approach for finding them
- Design new test cases to increase your mutation score by at least 10% for each class (Classes under test: Range.java, DataUtilities.java, etc.)



Assignment 4 – Part 2

- Install Selenium and SikuliX
- Follow instructions to
 - Design test cases for at least 2 functionalities (per student) of the selected websites
 - Automate your designed test cases using Selenium and add verification points to your scripts
- Compare Selenium with SikuliX